THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 1: **Chemical Reactions and Equations** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Magnesium ribbon burns in air with a dazzling white flame and forms a white powder. The powder is: 1

(a) MgO **(b)** $MgCl_2$ **(c)** $Mg(OH)_2$ **(d)** $MgCO_3$

2. The reaction $CaO + H_2O \longrightarrow Ca(OH)_2$ is: 1

(a) displacement and exothermic **(b)** decomposition and endothermic
(c) double displacement **(d)** combination and exothermic

3. For $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$, the balanced coefficients (in order) are: 1

(a) 3, 1, 1, 1 **(b)** 2, 3, 1, 3 **(c)** 3, 4, 1, 4 **(d)** 1, 1, 1, 1

4. A student states: “Decomposition reactions always need heat.” The correct response is: 1

(a) Correct
(b) Correct only for carbonates
(c) Incorrect — light or electricity can also be used
(d) Cannot be determined

5. For $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$, Student A calls it a displacement reaction; Student B calls it double displacement. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. White silver chloride turns grey in sunlight because it: 1

(a) absorbs moisture **(b)** decomposes into silver and chlorine
(c) combines with oxygen **(d)** melts

7. $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2\,NaCl$ is an example of: 1

(a) combination **(b)** redox
(c) decomposition **(d)** double displacement (precipitation)

8. In $CuO + H_2 \longrightarrow Cu + H_2O$, the substance oxidised is: 1

(a) H_2 **(b)** H_2O **(c)** CuO **(d)** Cu

9. Chips manufacturers flush packets with a gas to prevent rancidity. The gas is: 1

(a) hydrogen **(b)** carbon dioxide **(c)** nitrogen **(d)** oxygen

10. In the electrolysis of water, the volume ratio of hydrogen to oxygen collected is: 1

(a) 2 : 1 **(b)** 1 : 2 **(c)** 3 : 1 **(d)** 1 : 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why should a magnesium ribbon be cleaned before burning it in air? 2

12. Write the balanced equation for the reaction of hydrogen with chlorine and name the type of reaction. 2

13. What is corrosion? Give one example. 2

14. A student says that respiration is an endothermic process. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Translate into balanced chemical equations: 3
(i) Hydrogen combines with nitrogen to form ammonia.
(ii) Hydrogen sulphide burns in air to give water and sulphur dioxide.
(iii) Barium chloride reacts with aluminium sulphate to give aluminium chloride and barium sulphate.

16. What do you observe when ferrous sulphate crystals are heated? Write the equation and name the type of reaction. 3

17. An iron nail is dipped in copper sulphate solution for 20 minutes. Write two observations, the equation and the type of reaction. 3

OR

Explain oxidation and reduction with one example each. 3

18. Lead nitrate is heated in a boiling tube. 3
(i) What is seen?
(ii) Write the balanced equation.
(iii) Name the type of reaction.

OR

Why do we paint iron articles? Why are chips packets flushed with nitrogen? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. What is a balanced chemical equation and why should equations be balanced? Balance the following and name the type of each: 5
(a) $\mathrm{Na + H_2O \longrightarrow NaOH + H_2}$
(b) $\mathrm{Pb(NO_3)_2 + KI \longrightarrow PbI_2 + KNO_3}$
(c) $\mathrm{Al + CuCl_2 \longrightarrow AlCl_3 + Cu}$

OR

Explain decomposition reactions in which energy is supplied as 5
(i) heat
(ii) light
(iii) electricity, with one equation each. Why is silver bromide used in black-and-white photography?

20. **Case study:** Rohit added zinc granules to dilute sulphuric acid in a test tube. 5
(i) Which gas is evolved?
(ii) How will he test it?
(iii) Write the equation.
(iv) Name the type of reaction.
(v) What would happen if he used copper instead of zinc?

OR

(i) What is rancidity? 5
(ii) Give two ways to prevent it.
(iii) What is the chemical name of rust?
(iv) Give two ways to prevent rusting.

END OF QUESTION PAPER — Total: 40 Marks